

ME2 - VCM on Raspberry Pi 5

Always listening locally · two-stage TinyDSCNN-48 · no ASR

Two-stage offline speech-to-action

01

**Mic + log-mel front
end**

16 kHz · 2.5 s · 40 × 251

02

→ **Binary wake gate**

2 classes · threshold 0.95

03

→ **TinyDSCNN-48 intent**

31 classes · INT8 ONNX

04

→ **Coded action**

Light · media · temp ·
weather

10,682 training rows

ME2 Spoken Command Dataset

202 live holdout clips

186 commands · 16 OOS

74.9 KiB total

Wake + intent INT8 ONNX

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Live Pi acoustic holdout · quick mode · 125 trials · INT8

36.7%

intent accuracy · 34.7%
macro F1

33.9%

93-command accuracy ·
32.2% macro F1

50%

false wake · 8/16 no-wake
controls

Live Acoustic Benchmark Results

Intent accuracy: 36.7% [95% CI: 28%–46%] · macro F1
34.7%
93-command accuracy: 33.9% [95% CI: 26%–43%] · macro F1
32.2%
Intent by voice: 16.3% real (n=49) · 53.3% synthetic (n=60)

OOS false accept 62.5% (10/16) · false reject 36.6%

Pi 5 Acoustic & Runtime

Fifine mic check passed: SNR 21.8 dB.
109 wake-word clips + 16 no-wake
controls.

Inference p95 9.3 ms · RTF p95 0.004 ·
max temperature 45.2 °C

Slot exact 88.0% (n=25). 40/109 no response; 49
extra fires. Response p95 6.99 s is provisional:
16/69 timings negative.